FA1 REFLECTION

Biology Laboratory Record

1. Title & Objective

- **Experiment Title:** Release of Carbon Dioxide in Respiration
- **Objective:** To prove experimentally that carbon dioxide is released during the process of cellular respiration in living organisms (seeds).

2. Scientific Principle & Background

- Cellular respiration is a fundamental metabolic process occurring in all living cells, where organic food molecules (such as glucose) are broken down to release chemical energy in the form of ATP.
- Aerobic respiration consumes oxygen and releases carbon dioxide (CO_2), water, and energy.
- In this setup, germinating (sprouting) seeds are specifically selected because they are in an active stage of cellular growth, meaning their metabolic rate and respiration are significantly higher than dormant, unsprouted seeds.

3. Materials Required

- Glass bottles (two sets)
- Germinating (sprouted) seeds and unsprouted (dry) seeds
- Small beakers
- Freshly prepared lime water ($Ca(OH)_2$)
- Corks or airtight lids

4. Step-by-Step Procedure

- **Sample Preparation:** Take two clean glass bottles. Place a handful of germinating seeds in the first bottle and an equal quantity of unsprouted seeds in the second bottle as a control.
- **Indicator Placement:** Take a small beaker, fill it three-fourths with freshly prepared lime water, and carefully place it inside each of the glass bottles alongside the seeds.

- **Sealing:** Close both glass bottles tightly using airtight corks or lids to ensure no atmospheric gases can leak in or out.
- **Incubation:** Keep both experimental sets undisturbed in a safe place for one to two days to allow respiration to take place and gases to accumulate.
- **Observation:** Carefully open the bottles after the stipulated time and observe any physical change in the appearance of the lime water in both sets.

5. Observations

- In both bottles, the clear lime water gradually turns into a milky white solution.
- However, the rate of change is markedly different: the lime water in the bottle containing sprouted seeds turns milky white much faster and more intensely than the lime water in the bottle with unsprouted seeds.

6. Inference & Scientific Conclusion

- Lime water ($Ca(OH)_2$) reacts with carbon dioxide (CO_2) to form an insoluble white precipitate of calcium carbonate ($CaCO_3$), which turns the solution milky:

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$$

- The faster turning of lime water in the sprouted seeds set confirms that actively respiring seeds release a greater volume of carbon dioxide compared to dormant seeds, which have a minimal metabolic rate.

7. Precautions & Learning Outcomes

- Ensure the bottles are completely airtight; any leakage of atmospheric air will dilute the accumulated CO_2 and affect results.
- Fresh lime water must be used for clear, accurate observations.
- This experiment successfully demonstrates that living tissues respire and release carbon dioxide as a metabolic waste product.